

Motion in 2D

Single Correct Type Questions

1. The initial speed of a projectile fired from ground is u . At the highest point during its motion, the speed of projectile is $\frac{\sqrt{3}}{2}u$. The time of flight of the projectile is:

[31 Jan, 2023 (Shift-I)]

(a) $\frac{u}{2g}$

(b) $\frac{u}{g}$

(c) $\frac{2u}{g}$

(d) $\frac{\sqrt{3}u}{g}$

2. The trajectory of projectile, projected from the ground is given by $y = x - \frac{x^2}{20}$. Where x and y are measured in meter. The maximum height attained by the projectile will be.

[8 April, 2023 (Shift-II)]

(a) 5 m

(b) $10\sqrt{2}$ m

(c) 200 m

(d) 10 m

3. For a body projected at an angle with the horizontal from the ground, choose the correct statement.

[1 Feb, 2023 (Shift-II)]

- (a) Gravitational potential energy is maximum at the highest point.
(b) The horizontal component of velocity is zero at highest point.
(c) The vertical component of momentum is maximum at the highest point.
(d) The kinetic energy (K.E.) is zero at the highest point of projectile motion.

4. A projectile is projected at 30° from horizontal with initial velocity 40 ms^{-1} . The velocity of the projectile at $t = 2$ s from the start will be:

(Given $g = 10 \text{ m/s}^2$)

[11 April, 2023 (Shift-II)]

(a) $20\sqrt{3} \text{ ms}^{-1}$

(b) $40\sqrt{3} \text{ ms}^{-1}$

(c) 20 ms^{-1}

(d) Zero

5. Two projectiles are projected at 30° and 60° with the horizontal with the same speed. The ratio of the maximum height attained by the two projectiles respectively is:

[10 April, 2023 (Shift-II)]

(a) $2 : \sqrt{3}$

(b) $\sqrt{3} : 1$

(c) $1 : 3$

(d) $1 : \sqrt{3}$

6. The maximum vertical height to which a man can throw a ball is 136 m. The maximum horizontal distance upto which he can throw the same ball is

[24 Jan, 2023 (Shift-I)]

(a) 192 m

(b) 136 m

(c) 272 m

(d) 68 m

7. A projectile is launched at an angle ' α ' with the horizontal with a velocity 20 ms. After 10 s, its inclination with horizontal is ' β '. The value of $\tan \beta$ will be: ($g = 10 \text{ ms}^{-2}$).

[27 June, 2022 (Shift-I)]

(a) $\tan \alpha + 5 \sec \alpha$

(b) $\tan \alpha - 5 \sec \alpha$

(c) $2 \tan \alpha - 5 \sec \alpha$

(d) $2 \tan \alpha + 5 \sec \alpha$

8. A body of mass 10 kg is projected at an angle of 45° with the horizontal. The trajectory of the body is observed to pass through a point (20, 10). If T is the time of flight, then its momentum vector, at time $t = \frac{T}{\sqrt{2}}$ is _____

[Take $g = 10 \text{ m/s}^2$]

[27 July, 2022 (Shift-II)]

- (a) $100\hat{i} + (100\sqrt{2} - 200)\hat{j}$
 (b) $100\sqrt{2}\hat{i} + (100 - 200\sqrt{2})\hat{j}$
 (c) $100\hat{i} + (100 - 200\sqrt{2})\hat{j}$
 (d) $100\sqrt{2}\hat{i} + (100\sqrt{2} - 200)\hat{j}$

9. A projectile is projected with velocity of 25 m/s at an angle θ with the horizontal. After t seconds its inclination with horizontal becomes zero. If R represents horizontal range of the projectile, the value of θ will be:

[Use $g = 10 \text{ m/s}^2$]

[24 June, 2022 (Shift-I)]

- (a) $\frac{1}{2}\sin^{-1}\left(\frac{5t^2}{4R}\right)$ (b) $\frac{1}{2}\sin^{-1}\left(\frac{4R}{5t^2}\right)$
 (c) $\tan^{-1}\left(\frac{4t^2}{5R}\right)$ (d) $\cot^{-1}\left(\frac{R}{20t^2}\right)$

10. A person can throw a ball upto a maximum range of 100 m. How high above the ground he can throw the same ball?

[29 June, 2022 (Shift-II)]

- (a) 25 m (b) 50 m
 (c) 100 m (d) 200 m

11. A body is projected at $t = 0$ with a velocity 10 ms^{-1} at an angle of 60° with the horizontal. The radius of curvature of its trajectory at $t = 1 \text{ s}$ is R . Neglecting air resistance and taking acceleration due to gravity $g = 10 \text{ ms}^{-2}$ the radius of R is:

[11 Jan, 2019 (Shift-I)]

- (a) 10.3 m (b) 2.8 m
 (c) 2.5 m (d) 5.1 m

12. A shell is fired from a fixed artillery gun with an initial speed u such that it hits the target on the ground at a distance R from it. If t_1 and t_2 are the values of the time taken by it to hit the target in two possible ways, the product $t_1 t_2$ is:

[12 April, 2019 (Shift-I)]

- (a) R/g (b) $R/4g$
 (c) $2R/g$ (d) $R/2g$

13. The trajectory of a projectile near the surface of the earth is given as $y = 2x - 9x^2$. If it were launched at an angle θ_0 with speed v_0 then ($g = 10 \text{ ms}^{-2}$)

[12 April, 2019 (Shift-I)]

- (a) $\theta_0 = \cos^{-1}\left(\frac{2}{\sqrt{5}}\right)$ and $v_0 = \frac{3}{5} \text{ ms}^{-1}$

(b) $\theta_0 = \sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$ and $v_0 = \frac{5}{3} \text{ ms}^{-1}$

(c) $\theta_0 = \sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$ and $v_0 = \frac{3}{5} \text{ ms}^{-1}$

(d) $\theta_0 = \cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$ and $v_0 = \frac{5}{3} \text{ ms}^{-1}$

14. A stone is projected at angle 30° to the horizontal. The ratio of kinetic energy of the stone at point of projection to its kinetic energy at the highest point of flight will be

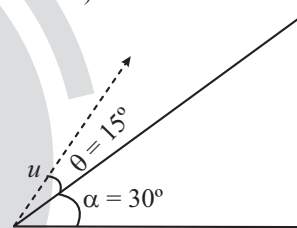
[29 Jan, 2023 (Shift-I)]

- (a) 1 : 2 (b) 1 : 4
 (c) 4 : 1 (d) 4 : 3

15. A plane is inclined at an angle $\alpha = 30^\circ$ with a respect to the horizontal. A particle is projected with a speed $u = 2 \text{ ms}^{-1}$ from the base of the plane, making an angle $\theta = 15^\circ$ with respect to the plane as shown in the figure. The distance from the base, at which the particle hits the plane is close to:

[10 April, 2019 (Shift-II)]

(Take $g = 10 \text{ ms}^{-2}$)



- (a) 14 cm (b) 20 cm
 (c) 18 cm (d) 26 cm

16. A passenger sitting in a train A moving at 90 km/h observes another train B moving in the opposite direction for 8 s. If the velocity of the train B is 54 km/h, then length of train B is:

[13 April, 2023 (Shift-II)]

- (a) 80 m (b) 200 m
 (c) 120 m (d) 320 m

17. Two buses P and Q start from a point at the same time and move in a straight line and their positions are represented by $X_P(t) = \alpha t + \beta t^2$ and $X_Q(t) = ft - t^2$. At what time, both the buses have same velocity?

[25 June, 2022 (Shift-II)]

(a) $\frac{\alpha - f}{1 + \beta}$

(b) $\frac{\alpha + f}{2(\beta - 1)}$

(c) $\frac{\alpha + f}{2(1 + \beta)}$

(d) $\frac{f - \alpha}{2(1 + \beta)}$

18. Train A and train B are running on parallel tracks in opposite directions with speeds of 36 km/hour and 72 km/hour, respectively, A person is walking in train A in the direction opposite to its motion with a speed of 1.8 km/hour. Speed (in ms^{-1}) of this person as observed from train B will be close to: (take the distance between the tracks as negligible) [2 Sep, 2020 (Shift-I)]

- (a) 28.5 ms^{-1} (b) 31.5 ms^{-1}
(c) 30.5 ms^{-1} (d) 29.5 ms^{-1}

19. A particle is moving with a velocity $\vec{v} = K(y\hat{i} + x\hat{j})$, where K is a constant. The general equation for its path is:

[9 Jan, 2019 (Shift-I)]

- (a) $y = x^2 + \text{constant}$
(b) $y^2 = x + \text{constant}$
(c) $y^2 = x^2 + \text{constant}$
(d) $xy = \text{constant}$

20. The stream of a river is flowing with a speed of 2 km/h. A swimmer can swim at a speed of 4 km/h. What should be the direction of the swimmer with respect to the flow of the river to cross the river straight?

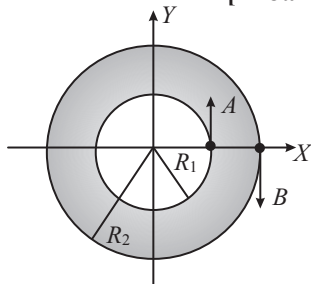
[9 April, 2019 (Shift-I)]

- (a) 60° (b) 150°
(c) 90° (d) 120°

21. Two projectile thrown at 30° and 45° with the horizontal respectively, reach the maximum height in same time. The ratio of their initial velocities is [26 July, 2022 (Shift-I)]

- (a) $1:\sqrt{2}$ (b) $2:1$
(c) $\sqrt{2}:1$ (d) $1:2$

22. Two particles A, B are moving on two concentric circles of radii R_1 and R_2 with equal angular speed ω . At $t = 0$, their positions and direction of motion are shown in the figure: [12 Jan, 2019 (Shift-II)]



The relative velocity $\vec{v}_A - \vec{v}_B$ at $t = \frac{\pi}{2\omega}$ is given by:

- (a) $\omega(R_1 + R_2)\hat{i}$
(b) $-\omega(R_1 + R_2)\hat{i}$
(c) $\omega(R_2 - R_1)\hat{i}$
(d) $\omega(R_1 - R_2)\hat{i}$

Integer Type Questions

23. If the initial velocity in horizontal direction of a projectile is unit vector $\hat{i}$ and the equation of trajectory is $y = 5x(1 - x)$. The y component vector of the initial velocity is _____ $\hat{j}$.

(Take $g = 10 \text{ m/s}^2$)

[26 July, 2022 (Shift-I)]

24. An object is projected in the air with initial velocity u at an angle θ . The projectile motion is such that the horizontal range R , is maximum. Another object is projected in the air with a horizontal range half of the range of first object. The initial velocity remains same in both the case. The value of the angle of projection, at which the second object is projected, will be _____ degree.

[29 July, 2022 (Shift-I)]

25. If the initial velocity in horizontal direction of a projectile is unit vector $\hat{i}$ and the equation of trajectory is $y = 5x(1 - x)$.

The y component vector of the initial velocity is _____ $\hat{j}$

(Take $g = 10 \text{ m/s}^2$)

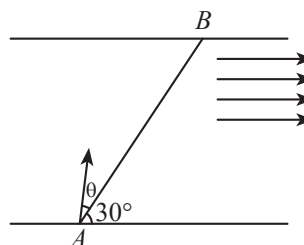
[JEE Adv, 2022]

26. A fighter jet is flying horizontally at a certain with a speed of 200 ms^{-1} . When it passes directly overhead ad anti-aircraft gun, a bullet is fired from the gun, at an angle θ with the horizontal, to hit the jet. If the bullet speed is 400 m/s , the value of θ will be _____.

[26 June, 2022 (Shift-I)]

27. A swimmer wants to cross a river from point A to point B. Line AB makes an angle of 30° with the flow of river. Magnitude of velocity of the swimmer is same as that of the river. The angle θ with the line AB should be _____ $^\circ$, so that the swimmer reaches point B.

[27 July, 2021 (Shift-II)]



28. A particle is moving along the x -axis with its coordinate with time ' t ' given by $x(t) = 10 + 8t - 3t^2$. Another particle is moving along the y -axis with its coordinate as a function of time given by $y(t) = 5 - 8t^3$. At $t = 1$ s, the speed of the second particle as measured in the frame of the first particle is given as $\sqrt{v}$. Then v (in m/s) is

[8 Jan, 2020 (Shift-I)]

29. A ball of mass m is thrown vertically upward. Another ball of mass $2m$ is thrown at an angle θ with the vertical. Both the balls stay in air for the same period of time. The ratio of the heights attained by the two balls respectively is $\frac{1}{x}$. The value of x is _____.

[JEE Adv, 2022]

30. An object is projected in the air with initial velocity u at an angle θ . The projectile motion is such that the horizontal range R , is maximum. Another object is projected in the air with a horizontal range half of the range of first object. The initial velocity remains same in both the case. The value of the angle of projection, at which the second object is projected, will be _____ degree. (Mark the smallest angle possible)

[JEE Adv, 2022]



ANSWER KEY

- | | | | | | | | | | |
|---------|---------|---------|--------------|---------|----------|----------|-----------|---------|----------|
| 1. (b) | 2. (a) | 3. (a) | 4. (a) | 5. (c) | 6. (c) | 7. (b) | 8. (d) | 9. (d) | 10. (b) |
| 11. (b) | 12. (c) | 13. (d) | 14. (d) | 15. (b) | 16. (d) | 17. (d) | 18. (d) | 19. (c) | 20. (d) |
| 21. (c) | 22. (c) | 23. [5] | 24. [15or75] | 25. [5] | 26. [60] | 27. [30] | 28. [580] | 29. [1] | 30. [15] |

EXPLANATIONS

1. (b) Velocity of the projectile at the highest point
 $= u \cos \theta$

$$u \cos \theta = \frac{\sqrt{3}u}{2}$$

$$\Rightarrow \theta = 30^\circ$$

$$\text{Time of flight } T = \frac{2u \sin 30^\circ}{g} = \frac{u}{g}$$

2. (a) $y = x - \frac{x^2}{20}$

For maximum height,

$$\frac{dy}{dx} = 0 \Rightarrow \frac{d}{dx} \left(x - \frac{x^2}{20} \right) = 0$$

$$1 - \frac{2x}{20} = 0$$

$$x = 10$$

$$\text{So, } y_{\max} = 10 - \frac{100}{20} = 5 \text{ m}$$

3. (a) At the highest point

Vertical component of the velocity, $v_y = 0$

Horizontal component of the velocity

$$v_x = u_x = u \cos \theta$$

Gravitational potential energy

$$U_g = mgh, \text{ it is maximum at } H_{\max}.$$

4. (a) Horizontal component of velocity, $v_x = u \cos 30^\circ$

$$= 40 \times \frac{\sqrt{3}}{2} = 20\sqrt{3} \text{ m/s}$$

Vertical component of velocity, $v_y = u \sin 30^\circ$

$$= 40 \times \frac{1}{2} = 20 \text{ m/s}$$

$$\text{At } t = 2 \text{ s, } v_x = u_x = 20\sqrt{3}$$

$$v_y = u_y - gt = 20 - 10 \times 2 = 0$$

$$\therefore v = \sqrt{v_x^2 + v_y^2} = 20\sqrt{3} \text{ ms}^{-1}$$

5. (c) Maximum height $H_{\max} = \frac{u^2 \sin^2 \theta}{2g} \Rightarrow H_{\max} \propto \sin^2 \theta$

$$\frac{H_1}{H_2} = \frac{\sin^2 \theta_1}{\sin^2 \theta_2} = \frac{\sin^2 30^\circ}{\sin^2 60^\circ} = \frac{(1/2)^2}{(\sqrt{3}/2)^2} = \frac{1}{3}$$

6. (c) Maximum vertical height is given by

$$H_{\max} = \frac{v^2}{2g} = 136 \text{ m}$$

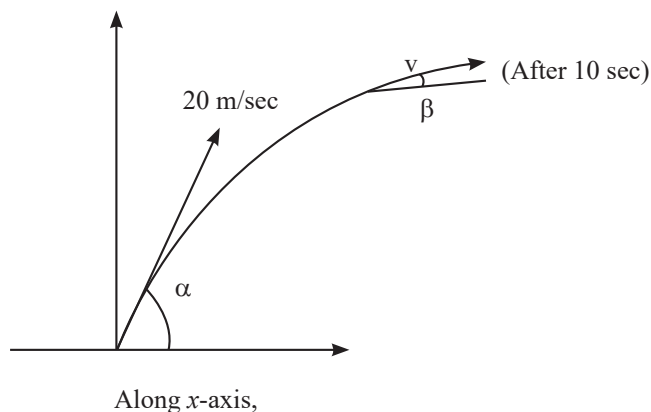
and maximum horizontal range is

$$R_{\max} = \frac{v^2}{g} = 2H_{\max}$$

$$= 2(136)$$

$$\Rightarrow R_{\max} = 272 \text{ m}$$

7. (b) Through out the motion, the horizontal speed will remain constant, therefore,



$$u_x = v_x = 20 \cos \alpha$$

Along y-axis

$$u_y = 20 \sin \alpha,$$

Applying Kinematic eqn.

$$v_y = u_y + a_y t$$

$$v_y = 20 \sin \alpha - 10 \times 10$$

$$\tan \beta = \frac{v_y}{v_x} = \frac{20 \sin \alpha - 100}{20 \cos \alpha}$$

$$\Rightarrow \tan \beta = \tan \alpha - 5 \sec \alpha$$

...(i)

...(ii)

$$8. (d) \quad y = x - \frac{10x^2}{2u^2 \left(\frac{1}{2}\right)}$$

$$\Rightarrow 10 = 20 - \frac{(10)(400)}{u^2} \Rightarrow u = 20 \text{ m/s}$$

$$t = \frac{T}{\sqrt{2}} = \frac{(2)(20) \times \sin 45^\circ}{\sqrt{2}(10)} = 2\sqrt{s}$$

$$\vec{v} = u \cos \theta \hat{i} + (u \sin \theta - gt) \hat{j} = 10\sqrt{2} \hat{i} + (10\sqrt{2} - 10(2)) \hat{j}$$

$$\text{Momentum } \vec{p} = M\vec{v} = 100\sqrt{2} \hat{i} + (100\sqrt{2} - 200) \hat{j}$$

$$9. (d) \quad R = \frac{V^2 (2 \sin \theta \cos \theta)}{g}$$

$$t = \frac{V \sin \theta}{g} \Rightarrow V = \frac{gt}{\sin \theta}$$

$$\Rightarrow R = \frac{g^2 t^2}{\sin^2 \theta} \cdot \frac{2 \sin \theta \cos \theta}{g}$$

$$\tan \theta = \frac{2gt^2}{R} = \frac{20t^2}{R}$$

$$\cot \theta = \frac{R}{20t^2}$$

10. (b) According to question, we can write

$$R = \frac{u^2 \sin 2\theta}{g}$$

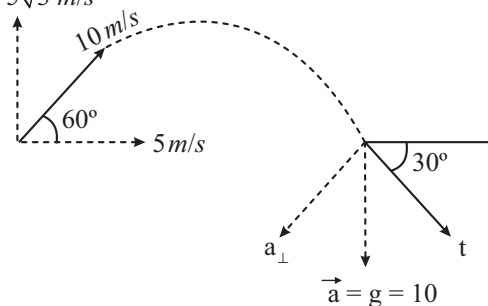
For maximum range, ($R_{\max}$)

$$\theta = 45^\circ.$$

$$R_{\max} = \frac{u^2}{g} \Rightarrow u^2 = gR_{\max}$$

$$\Rightarrow H_{\max} = \frac{u^2}{2g} = \frac{gR_{\max}}{2g} = \frac{R_{\max}}{2} = \frac{100}{2} = 50 \text{ m}$$

11. (b) $5\sqrt{3} \text{ m/s}$



At $t = 1$

$$u_x = 5, u_y = 5\sqrt{3}$$

$$v_y = 5\sqrt{3} - 10; v_x = 5$$

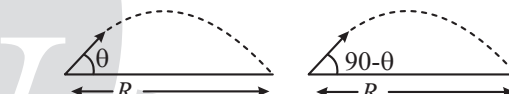
Angle made by velocity vector at $t = 1$ sec.

$$|\tan \alpha| = \left| \frac{v_y}{v_x} \right| = |2 - \sqrt{3}|$$

$$\alpha = 15^\circ$$

$$\therefore R = \frac{5^2 + (10 - 5\sqrt{3})^2}{10 \cos \alpha} = \frac{200 - 100\sqrt{3}}{10 \times 0.965} = 2.8 \text{ m}$$

12. (c) Range will be same for time t_1 and t_2 , so angles of projection will be ' θ ' and ' $90^\circ - \theta$ '



$$t_1 = \frac{2u \sin \theta}{g} \quad t_2 = \frac{2u \sin (90^\circ - \theta)}{g}$$

$$\text{and } R = \frac{u^2 \sin 2\theta}{g}$$

$$t_1 t_2 = \frac{4u^2 \sin \theta \cos \theta}{g^2} = \frac{2}{g} \left[\frac{2u^2 \sin \theta \cos \theta}{g} \right] = \frac{2R}{g}$$

13. (d) Equation of trajectory is given as

$$y = 2x - 9x^2 \quad \dots (i)$$

Comparing with equation:

$$y = x \tan \theta - \frac{g}{2u^2 \cos^2 \theta} \cdot x^2 \quad \dots (ii)$$

We get; $\tan \theta = 2$

$$\therefore \cos \theta = \frac{1}{\sqrt{5}}$$

$$\text{Also, } \frac{g}{2u^2 \cos^2 \theta} = 9$$

$$\Rightarrow \frac{10}{2 \times 9 \times \left(\frac{1}{\sqrt{5}}\right)^2} = u^2 \Rightarrow u^2 = \frac{25}{9} \Rightarrow u = \frac{5}{3} \text{ m/s}$$

14. (d) At the maximum height velocity = $u \cos \theta = u \cos 30^\circ$

$$\frac{KE_{\text{initial}}}{KE_{\text{top}}} = \frac{\frac{1}{2} M(u)^2}{\frac{1}{2} M(u \cos 30^\circ)^2} = \frac{4}{3}$$

15. (b) $t = \frac{2 \times 2 \sin 15^\circ}{g \cos 30^\circ}$

$$S = 2 \cos 15^\circ \times t - \frac{1}{2} g \sin 30^\circ t^2$$

Put values and solve: $S = 20 \text{ cm}$

16. (d) $\boxed{B} \xrightarrow{V_B} \leftarrow \xrightarrow{V_A} \boxed{A}$

Velocity of train A, $V_A = 90 \text{ km/h} = -2.5 \text{ m/s}$

$$\therefore 1 \text{ km/h} = \frac{5}{18} \text{ m/s}$$

Velocity of train B, $V_B = 54 \text{ km/h} = 15 \text{ m/s}$

Velocity of train B w.r.t. train A

$$\vec{V} = \vec{V}_B - \vec{V}_A = 15 - (-25) = 40 \text{ m/s}$$

$$\text{Time of crossing (t)} = \frac{\text{length of train (l)}}{\text{relative velocity (V)}}$$

$$(8) = \frac{l}{40} \Rightarrow l = 8 \times 40 = 320 \text{ m}$$

17. (d) $X_p(t) = \alpha t + \beta t^2$

$$V_p(t) = \frac{dX_p(t)}{dt} = \alpha + 2\beta t$$

$$X_Q(t) = ft - t^2$$

$$V_Q(t) = \frac{dX_Q(t)}{dt} = f - 2t$$

According to the question, (i) = (ii)

$$\alpha + 2\beta t = f - 2t \Rightarrow t(2\beta + 2) = f - \alpha$$

$$\therefore t = \frac{f - \alpha}{2(\beta + 1)}$$

18. (d) $V_A = 36 \text{ km/hr} = 10 \text{ m/s}$

$$V_B = -72 \text{ km/hr} = -20 \text{ m/s}$$

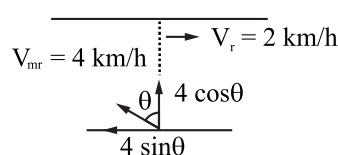
$$V_{MA} = -1.8 \text{ km/hr} = -0.5 \text{ m/s}$$

$$V_{\text{man}, B} = V_{\text{man}, A} + V_{A, B} = -0.5 + 10 - (-20) = 29.5 \text{ m/s}$$

19. (c) $\frac{dx}{dt} = y; \frac{dy}{dt} = x$

$$\frac{dx}{dy} = \frac{y}{x} \Rightarrow y^2 = x^2 + c$$

20. (d)



For swimmer to cross the river straight

$$\Rightarrow 4 \sin \theta = 2$$

$$\Rightarrow \sin \theta = \frac{1}{2} \Rightarrow \theta = 30^\circ$$

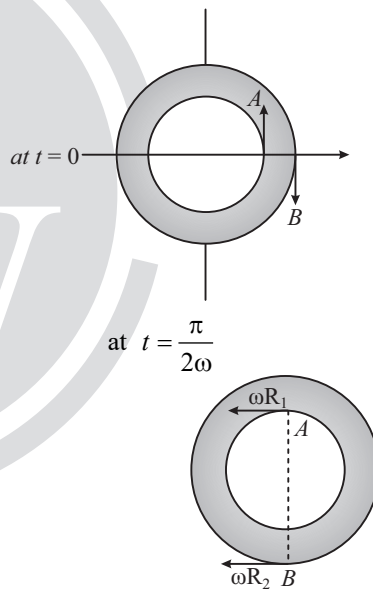
So, angle with direction of river flow = $90^\circ + \theta = 120^\circ$

21. (c) Time taken to reach maximum height

$$t = \frac{u \sin \theta}{g} \therefore \frac{u_1 \sin \theta_1}{g} = \frac{u_2 \sin \theta_2}{g}$$

$$\Rightarrow u_1 \sin 30^\circ = u_2 \sin 45^\circ \Rightarrow \frac{u_1}{u_2} = \frac{1/\sqrt{2}}{1/2} = \frac{\sqrt{2}}{1}$$

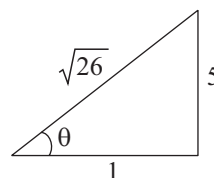
22. (c)



$$\vec{v}_A - \vec{v}_B = -\omega R_1 \hat{i} + \omega R_2 \hat{i} = \omega(R_2 - R_1) \hat{i}$$

23. [5] Given, $u_x = \hat{i}, |u_x| = 1$

$$y = x^5(1-x) = x \tan \theta \left(1 - \frac{x}{R}\right)$$



$$\tan \theta = 5, R = 1$$

$$\tan \theta = \frac{|4u_x|}{|4u_y|}$$

$$|4u_y| = |4u_x| \tan \theta = 1 \times 5 = 5 \text{ m/s}$$

$$24. [15 \text{ or } 75] \quad R = \frac{u^2 \sin(2 \times 45^\circ)}{g} = \frac{u^2}{g}$$

$$\frac{R}{2} = \frac{u^2}{2g} = \frac{u^2 \sin 2\theta}{g}$$

$$\Rightarrow \sin 2\theta = \frac{1}{2} \Rightarrow \theta = 15^\circ, 75^\circ$$

$$25. [5] \quad u_x = 1$$

$$y = 5x(1-x)$$

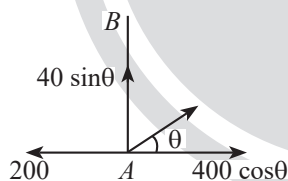
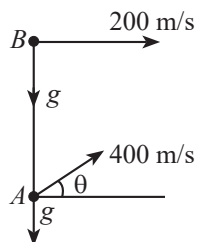
$$\frac{dy}{dt} = 5 \frac{dx}{dt} - 10x \frac{dx}{dt}$$

For initial y-component of velocity

$$u_y = \left(\frac{dy}{dt} \right)_{x=0} \Rightarrow 5(1) = 5$$

$$\vec{u}_y = 5\hat{j}$$

$$26. [60] \quad \text{If } A \text{ hits } B$$



Then relative velocity perpendicular to the line joining A to B will be zero.

$$\Rightarrow 400 \cos \theta = 200$$

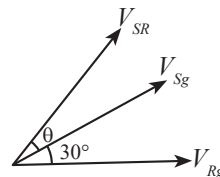
$$\cos \theta = \frac{1}{2}$$

$$\theta = 60^\circ$$

27. [30] Both velocity vector are of same magnitude. Therefore resultant would pass exactly midway through them.

So, $\theta = 30^\circ$

Hence, angle θ with the line AB is 30°



$$28. [580]$$

$$X_1 = -3t^2 + 8t + 10$$

$$\vec{v}_1 = (-6t + 8)\hat{i} = 2\hat{i}$$

$$Y_2 = 5 - 8t^3$$

$$\vec{v}_2 = -24t^2\hat{j}$$

$$\sqrt{v} = |\vec{v}_2 - \vec{v}_1| = |-24\hat{j} - 2\hat{i}|$$

$$\sqrt{v} = \sqrt{24^2 + 2^2}$$

$$v = 580 \text{ m/s}$$

$$29. [1] \quad \text{Time of flight is same}$$

$\Rightarrow$ Vertical component of velocity is same

$\Rightarrow H_{\max}$ is same

$$30. [15] \quad R_{\max} = \frac{u^2 \sin 2(45^\circ)}{g} = \frac{u^2}{g}$$

$$\frac{R}{2} = \frac{u^2}{2g} = \frac{u^2 \sin 2\theta}{g}$$

$$\sin 2\theta = \frac{1}{2}$$

$$2\theta = 30^\circ, 150^\circ$$

$$\theta = 15^\circ, 75^\circ$$